PRAKASH SWAMI

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Summary

Full-Stack Software Developer and Computer Engineering student with production experience shipping a Next.js and TypeScript SaaS platform. Skilled in React.js, Node.js, Flutter, PostgreSQL, and REST APIs, with hands-on work in authentication, payments, performance optimization, and AI/LLM integrations (Gemini, OpenAI, RAG).

Education

Vishwakarma Institute of Technology (VIT), Pune *2024 – 2028*
B.Tech in Computer Engineering | CGPA: 8.84/10
Relevant Coursework: Data Structures & Algorithms, DBMS, Operating Systems, OOP, Agile Software Development

Higher Secondary Certificate (HSC, 12th Grade) — 82.34% *2024*

Technical Skills

Languages: JavaScript, TypeScript, Python, Dart, C++, SQL, Solidity
Frontend & Mobile: React.js, Next.js, Flutter, React Native, Tailwind CSS, Vite, HTML5, CSS3
Backend & Databases: Node.js, Express.js, FastAPI, REST APIs, PostgreSQL, Prisma, Supabase, Firebase, Redis
AI/ML: LangChain, OpenAI API, Gemini API, RAG, ChromaDB, YOLOv8, TensorFlow Lite, Computer Vision
Cloud & Tools: Git, GitHub, AWS S3, Vercel, Stripe, Clerk, Hardhat, IPFS, Postman, Agile/Scrum

Experience

Software Development Engineer Intern — Career Craftly *May 2026 – Present*
Project: The GP Edge | *Tech Stack: Next.js, TypeScript, Prisma, PostgreSQL, Clerk, Stripe, Gemini API*

- Built a full-stack SaaS exam-prep and billing platform for Australian GP registrars across 38 pages using Next.js App Router, Server Actions, Prisma, and Neon PostgreSQL, with a learner dashboard and a full admin CMS.
- Implemented Clerk authentication with role-based access control and Stripe subscription billing with webhook-driven access management, plus REST APIs for quizzes, mock tests, content, and notifications.
- Engineered an AI-powered Medicare (MBS) billing explorer using Gemini structured-JSON parsing and database caching, turning raw government billing notes into searchable items, pathways, and compliance checklists.
- Optimized page performance by moving access checks to the server, replacing raw table scans with SQL aggregations, and removing redundant queries, cutting exam-prep LCP by 38%, navigation time by 64%, and payload by 94%.

Projects

MediChain — Blockchain-Audited Medical Records Platform
Tech Stack: Flutter, React.js, Node.js, Express.js, Solidity, Hardhat, Polygon, Supabase, IPFS

- Built a patient-owned medical records platform with a Flutter mobile app, React web portal, and Node.js/Express REST API, supporting record sharing with expiry, revocation, appointment booking, and Google sign-in.
- Secured files with AES-256 encryption, SHA-256 integrity checks, IPFS storage, and Supabase row-level security, plus a watermarked, screenshot-blocked viewer.
- Developed Solidity smart contracts on Polygon for time-bound access grants and an immutable audit log, with an ERC-2771 gasless relayer and EIP-712 signatures.

SolarSense AR — AI-Powered Rooftop Solar Assessment App
Tech Stack: Flutter, Dart, TensorFlow Lite, YOLOv8, PVGIS API, FastAPI, Python, OpenAI API

- Built a Flutter app that produces rooftop solar feasibility reports in under 2 minutes via an AR camera scan.
- Deployed an on-device YOLOv8 Nano (TensorFlow Lite) model to detect rooftop obstacles and compute usable roof area, running inference concurrently with PVGIS solar-irradiance API calls.
- Implemented system sizing, ROI, payback, and PM Surya Ghar subsidy calculations with a 16-section PDF report, backed by a FastAPI service with LLM-generated insights.

PaperTrail — RAG Knowledge Retrieval System
Tech Stack: Python, OpenAI GPT-4o, ChromaDB, Sentence-Transformers, Streamlit, PyMuPDF

- Built a Retrieval-Augmented Generation (RAG) system that ingests PDF, Word, Excel, CSV, and text files, using row-level chunking for spreadsheets and title-based chunking for documents.
- Implemented semantic search with Sentence-Transformer embeddings in ChromaDB and grounded GPT-4o prompts that return answers with file, page, and row citations, reducing hallucinations.

Certifications

- AWS Cloud Technology Consultant Professional Certificate — Amazon Web Services, Coursera
- Complete Full-Stack Web Development Bootcamp — The App Brewery, Udemy
- AI Agents: Automation & Business with LangChain & LLM Apps — Udemy

Achievements & Leadership

- Finalist at CanHack International and 4 national-level hackathons.
- Solved 470+ Data Structures and Algorithms problems on LeetCode.
- Led development of the official ICON club website, coordinating the web team and managing event updates.